SOFTWARE ENGINEERING MINI-PROJECT · TEAM 07

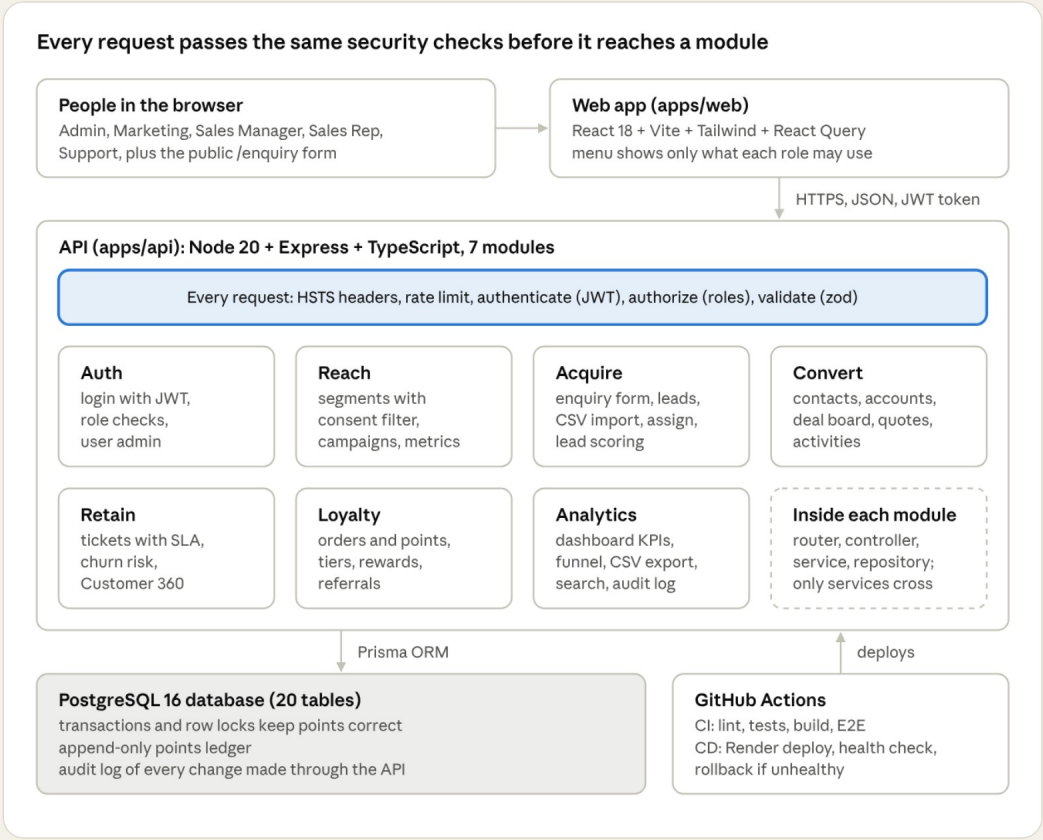
CRM-RACRL

One web app that takes a person from a marketing campaign to a lead, a paying customer and a loyal repeat customer.



Balla Santhosh · Dhriti Kamani · Disha Lokesh · Diya Agarwal — PES University RR · Section B · v1.0.0

One web app, one API, one database



Modular monolith

7 modules, each router → controller → service → repository. Modules talk only through services.

Security on every request

JWT login and role checks on every route. A test fails the build if any route skips them.

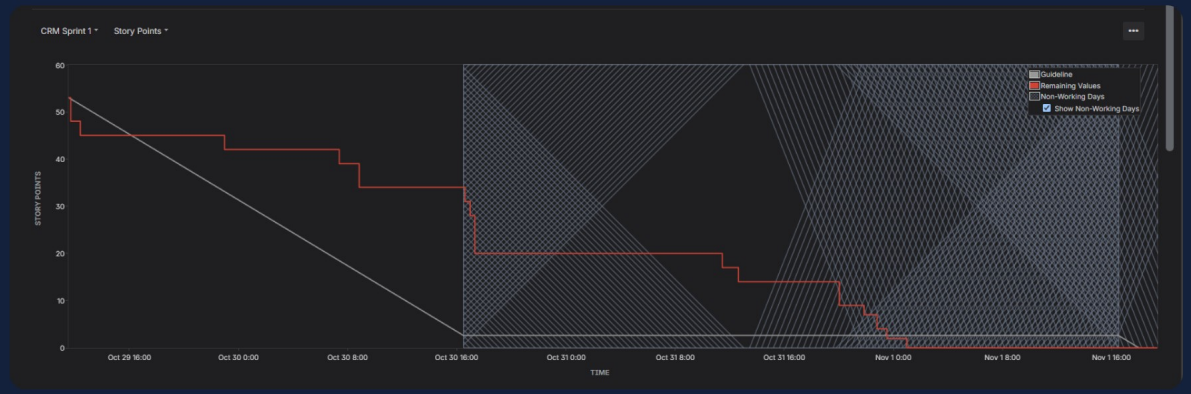
Money and points stay correct

Transactions and row locks in PostgreSQL, an append-only points ledger, totals in paise.

React 18 + Vite · Node 20 + Express + TypeScript · PostgreSQL 16 + Prisma · GitHub Actions CI/CD · Render

Two sprints, every committed point delivered

Sprint 1 · 29 Oct – 1 Nov · 53 points



Sprint 2 · 2 – 7 Nov · 54 points



107 / 107

story points done, 33 stories

39 in 2 days

of Sprint 2's 54 points, ahead of plan

37 PRs

each reviewed and tagged with a Jira key

A story counted as Done only after its PR merged. Jira's guideline also treats weekends as non-working, so our line sits above it.

Every merge is tested, then deployed

409
API tests on a real database

112 + 18
web tests + end-to-end browser tests

98%
API line coverage (build fails under 70%)

50 ms
p95 API latency, 50 users (target 500 ms)

100
Lighthouse accessibility, 0 axe issues

360 px
phone width, no sideways scroll



Next: a Convert button on the Leads page, a real email provider for campaigns, and smaller stories so the burndown moves daily.